



CASE STUDY

Police Station | Salt, Spain

POLICE STATION

Spatial Intelligence

PHASE 02 | GROUND FLOOR ANALYSIS

// 01 **ANALYSIS GRAPH**

> spatial substrate

// 04 **SHORTEST PATH**

> movement efficiency

// 07 **DEGREE CENTRALITY**

> junction connectivity

// 02 **CLOSENESS CENTRALITY**

> corridor integration

// 05 **COMMUNITY PARTITION**

> functional zones

// 08 **ISOVIST VIEWPOINTS**

> visibility sampling

// 03 **BETWEENNESS CENTRALITY**

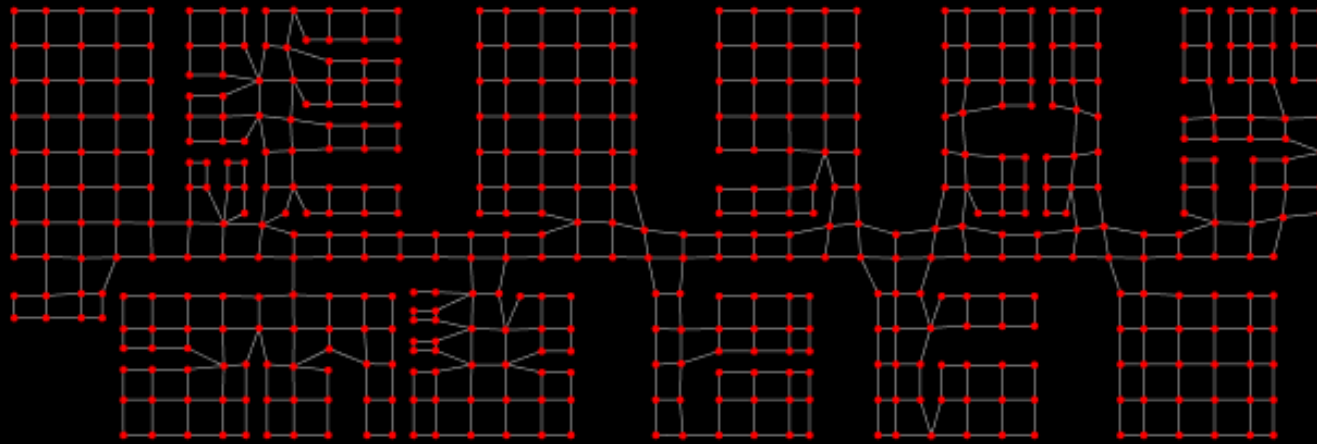
> routing chokepoint

// 06 **COMMUNITY GRAPH**

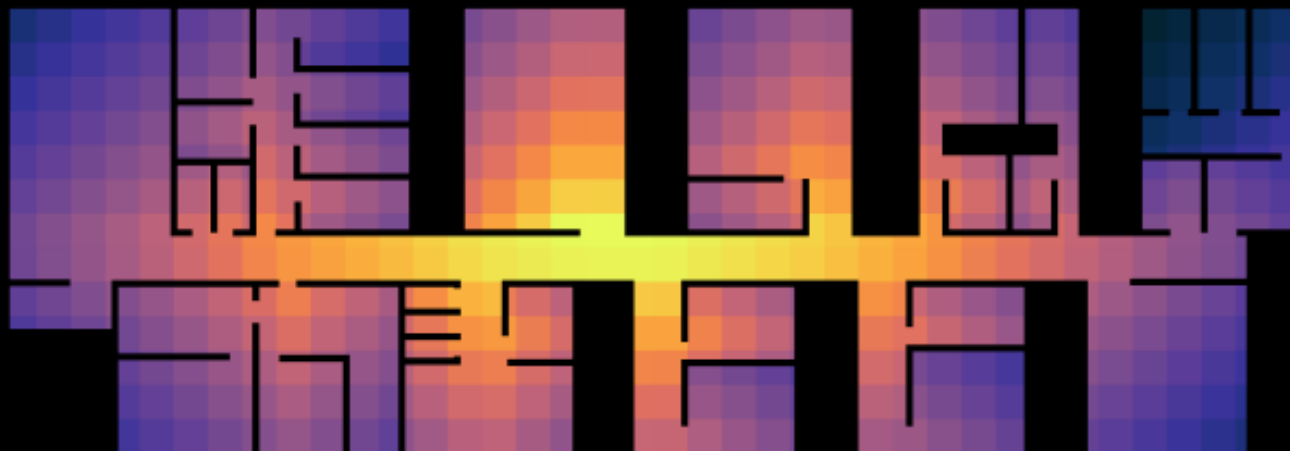
> inter-zone topology

// 09 **ISOVIST VISIBILITY MAP**

> surveillance exposure



- > 2x2m grid discretizes the ground floor into nodes and edges
- > shared cell boundaries become graph connections
- > spatial proximity encoded as topological structure



// 02

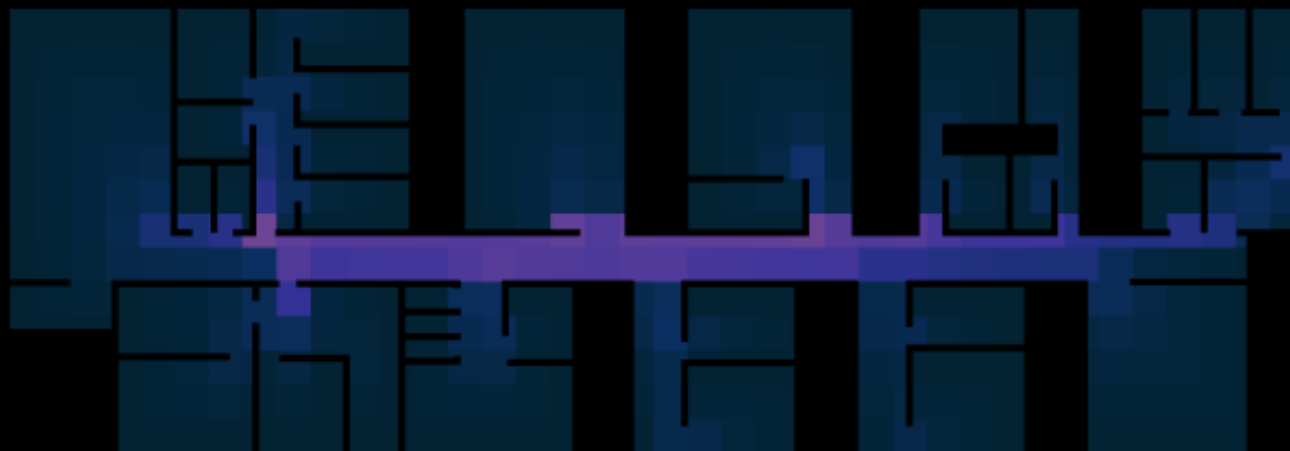
CLOSENESS CENTRALITY

| INTEGRATION

- > corridor is the shallowest space, reachable in the fewest steps
- > peripheral wings read as topologically isolated from the centre
- > warm tones = high integration; cool tones = topological depth

Closeness Centrality





// 03

BETWEENNESS CENTRALITY

| CHOICE

- > single east-west spine concentrates almost all routing choice
- > nearly every shortest path between spaces crosses the corridor
- > one structural bottleneck connects the entire building

Betweenness Centrality



- > cross-building traversal computed from NW to SE corner
- > graph path follows cell boundaries: 90.65 units
- > geometric straightening cuts across open space: 80.41 units (-11.3%)

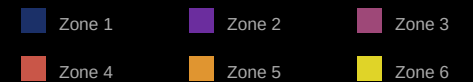
— Straightened | 80.41 units (-11.3%)

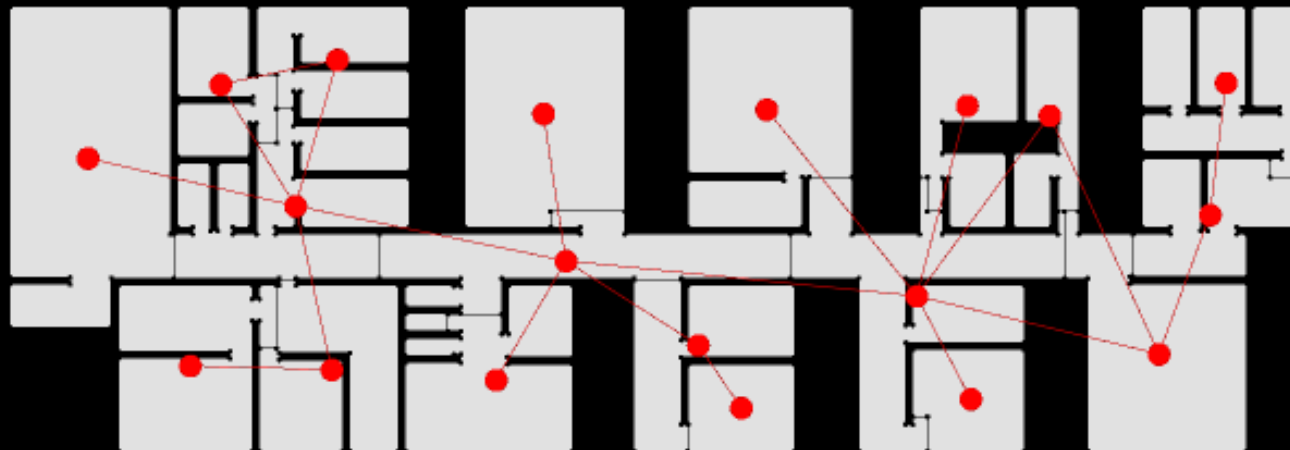


// 05

COMMUNITY PARTITION

- > 6 spatially coherent zones emerge from the graph structure
- > each cluster is more connected internally than to its neighbours
- > zones align with functional wings: offices, cells, circulation, public

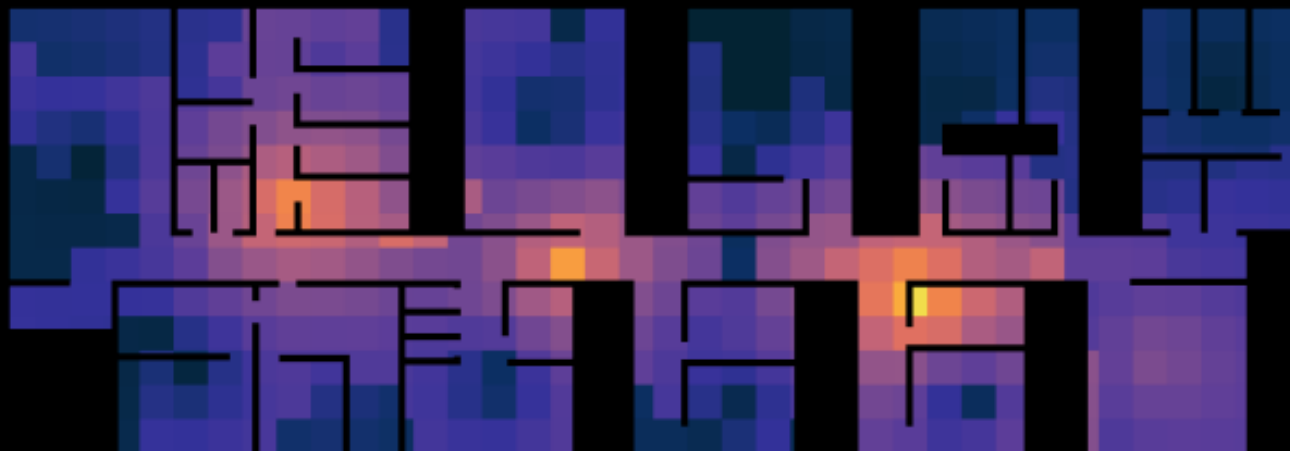




// 06

COMMUNITY GRAPH

- > communities collapsed to single nodes reveal inter-zone topology
- > corridor community holds the highest degree, acting as central hub
- > its removal would fragment the entire building into isolated clusters



// 07

DEGREE CENTRALITY

| COMMUNITY SCALE

- > degree centrality interpolated from community nodes back to the grid
- > junction zones between corridor and wings score highest
- > reveals the structural pivots embedded in the floor plan geometry

Degree Centrality





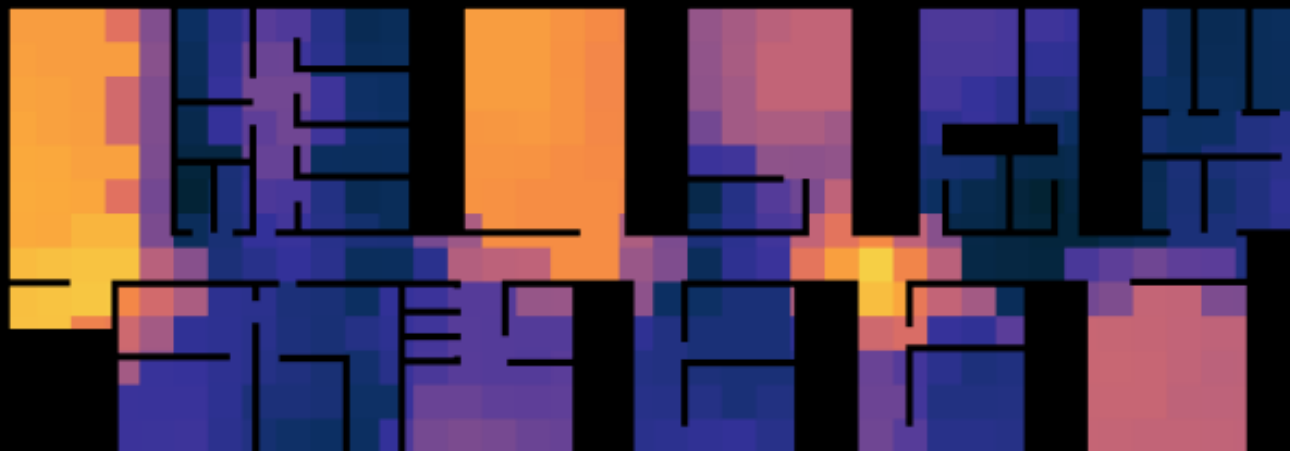
// 08

ISOVIST VIEWPOINTS

- > 61 viewpoints sampled at 5m intervals across the floor plan
- > each isovist polygon captures the total visible area from that point
- > visibility ranges from 6 to 55 grid cells, a 9x spread across the plan

● Viewpoint (61 total)

▴ Isovist (visibility polygon)



- > visibility scores interpolated across the full floor plan grid
- > corridor junctions and open spaces register the highest exposure
- > subdivided rooms and dead-end wings are effectively hidden

